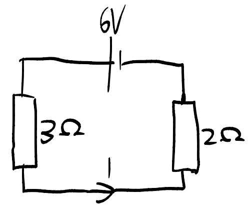
Kirchhoff's Laws PPS

1

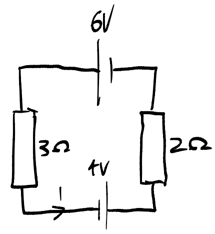


$R = \frac{V}{I}$

$5 = \frac{6}{I}$

$I = 1.2A$

2

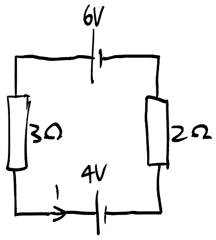


$R = \frac{V}{I}$

$5 = \frac{10}{I}$

$I = 2A$

3

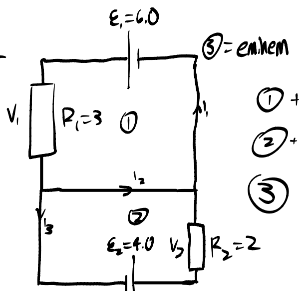


$R = \frac{V}{I}$

$5 = \frac{2}{I}$

$I = 0.4A$

4



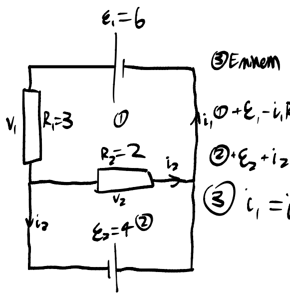
③ = eminem

① $+\varepsilon_1 - i_1R_1 = 0 \to i_1 = 2A$

② $+\varepsilon_2 - i_3R_2 = 0 \to i_3 = 2A$

③ $i_1 = i_2 + i_3 \to i_2 = 0A$

5



③ Eminem

① $+\varepsilon_1 - i_1R_1 - i_2R_2 = 0$

② $+\varepsilon_2 + i_2R_2 = 0 \to i_2 = -2A$ (2A →)

③ $i_1 = i_2 + i_3$

From ①: $+\varepsilon_1 - i_1R_1 - i_2R_2 = 0$

$$i_1 = \frac{i_2R_2 - \varepsilon_1}{R_1}$$

$$= 2A$$

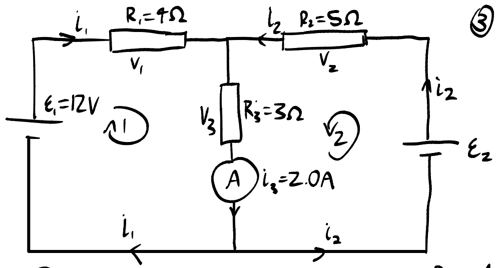
③ $i_1 = i_2 + i_3$

$2A = 2A + i_3$

$i_3 = 0A$

6 is the same as 5.

Kirchhoff's Laws PPS cont.



③ eminem

⓪ $i_1+i_2=i_3$

① $+\mathcal{E}_1-i_1R_1-i_3R_3=0$

② $+\mathcal{E}_2-i_2R_2-i_3R_3=0$

③ $+\mathcal{E}_1-i_1R_1+i_2R_2-\mathcal{E}_2=0$

$R_1=4$ $\quad \mathcal{E}_1=12$

$R_2=5$ $\quad \mathcal{E}_2=?$

$R_3=3$

$i_3=2$

I $\quad i_1=\frac{\mathcal{E}_1-i_3R_3}{R_1}$ $\quad$ ①

$i_2=i_3-i_1$ $\quad$ ⓪

II $\quad =i_3-\frac{\mathcal{E}_1-i_3R_3}{R_1}$ $\quad$ (I)

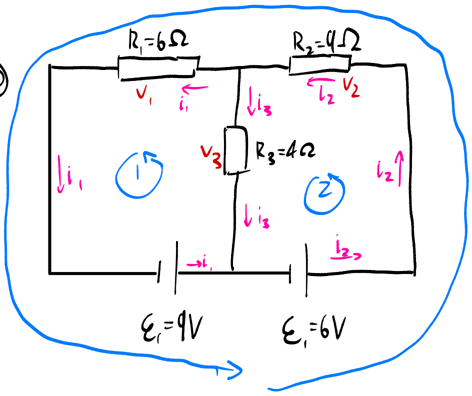
$\mathcal{E}_2=i_2R_2+i_3R_3$ $\quad$ ②

$=R_2\left(i_3-\frac{\mathcal{E}_1-i_3R_3}{R_1}\right)+i_3R_3$ $\quad$ (II)

$=i_3R_2+i_3R_3-\frac{R_2\mathcal{E}_1-R_2i_3R_3}{R_1}$

$=8.5$

D)



I) Need K_1 and K_2

A) Apply K_1 and K_2 to (D) step

K_1 @ junction: $\boxed{i_1+i_3=i_2}$ (I)

For loop 1:

$+\mathcal{E}_1-(-i_3R_3)-i_1R_1=0$

$\boxed{\mathcal{E}_1=i_1R_1-i_3R_3}$ (II)

For loop 2:

$+\mathcal{E}_2-i_2R_2-i_3R_3=0$

$\boxed{\mathcal{E}_2=i_2R_2+i_3R_3}$ (III)

Sub (I) into (III)

$\mathcal{E}_2=(i_1+i_3)R_2+i_3R_3$

$i_3R_3=i_1R_1-\mathcal{E}_1$ $\quad$ (from (II))

$\Rightarrow \mathcal{E}_2=(i_1+i_3)R_2+i_1R_1-\mathcal{E}_1$

$i_3=\frac{i_1R_1-\mathcal{E}_1}{R_3}$

$\mathcal{E}_2=\left(i_1+\frac{i_1R_1-\mathcal{E}_1}{R_3}\right)R_2+i_1R_1-\mathcal{E}_1$

$\mathcal{E}_2=i_1R_2+i_1\frac{R_1R_2-\mathcal{E}_1R_2}{R_3}+i_1R_1-\mathcal{E}_1$

$=i_1\left(R_2+\frac{R_1R_2-\mathcal{E}_1R_2}{R_3}+R_1\right)-\mathcal{E}_1$

$\frac{\mathcal{E}_2+\mathcal{E}_1}{R_2+\frac{R_1R_2-\mathcal{E}_1R_2}{R_3}+R_1}=i_1$

$i_1=1.24$